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PYSPARK QUERY · PART 3 QUICK REFERENCE

Spark SQL · safe managed Delta summary

Same answers in SQL. Persist one new managed summary with mode("error").
Free Edition: serverless + Unity Catalog. Unique schema suffix per run.

1. Temp view + high-value SQL

```
valid.createOrReplaceTempView("valid_orders")
spark.sql("""SELECT order_id, customer, order_total
  FROM valid_orders WHERE order_total >= 25
  ORDER BY order_total DESC, order_id DESC""")
```

Actual: 108 and 104 at 30.00 each · total 60.00.

2. SQL aggregation (match DataFrame)

```
spark.sql("""SELECT region, category, COUNT(*) AS order_count,
  CAST(SUM(order_total) AS DECIMAL(12,2)) AS revenue
  FROM valid_orders GROUP BY region, category
  ORDER BY region, category""")
```

Actual: same 5 buckets · 6 orders · 142.50 · match assertion passed.

3. Unique destination

```
schema_fqn = f"workspace.bt_query_{uuid4().hex[:12]}"
spark.sql(f"CREATE SCHEMA {schema_fqn}")
table_fqn = f"{schema_fqn}.region_category_revenue"
```

Actual lesson: workspace.bt_query_cf911148fc3d.region_category_revenue

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mode(error) write + verify

4. Safe managed Delta write

```
(summary_df.write.format("delta")
  .mode("error")   # NOT overwrite / drop / append
  .saveAsTable(table_fqn))
# Run the destination cell ONCE.
```

mode("error") fails if the table already exists — that is the safety rail.

5. Verify persisted rows

```
stored = spark.table(table_fqn)
# Assert 5 expected buckets; gate: 5 / 6 / 142.50
# PASS: stored summary = 5 buckets / 6 orders / revenue 142.50
```

Persisted: E/Books 12 · N/Books 20 · N/Toys 60 · S/Toys 25 · W/Books 25.50.

Associate §3 Gold tables + §7 managed Delta (Aug 2026). No exam guarantee.
Independent education. Fictional-data project. Part 3 of 3.